Term Schedule Visualizer

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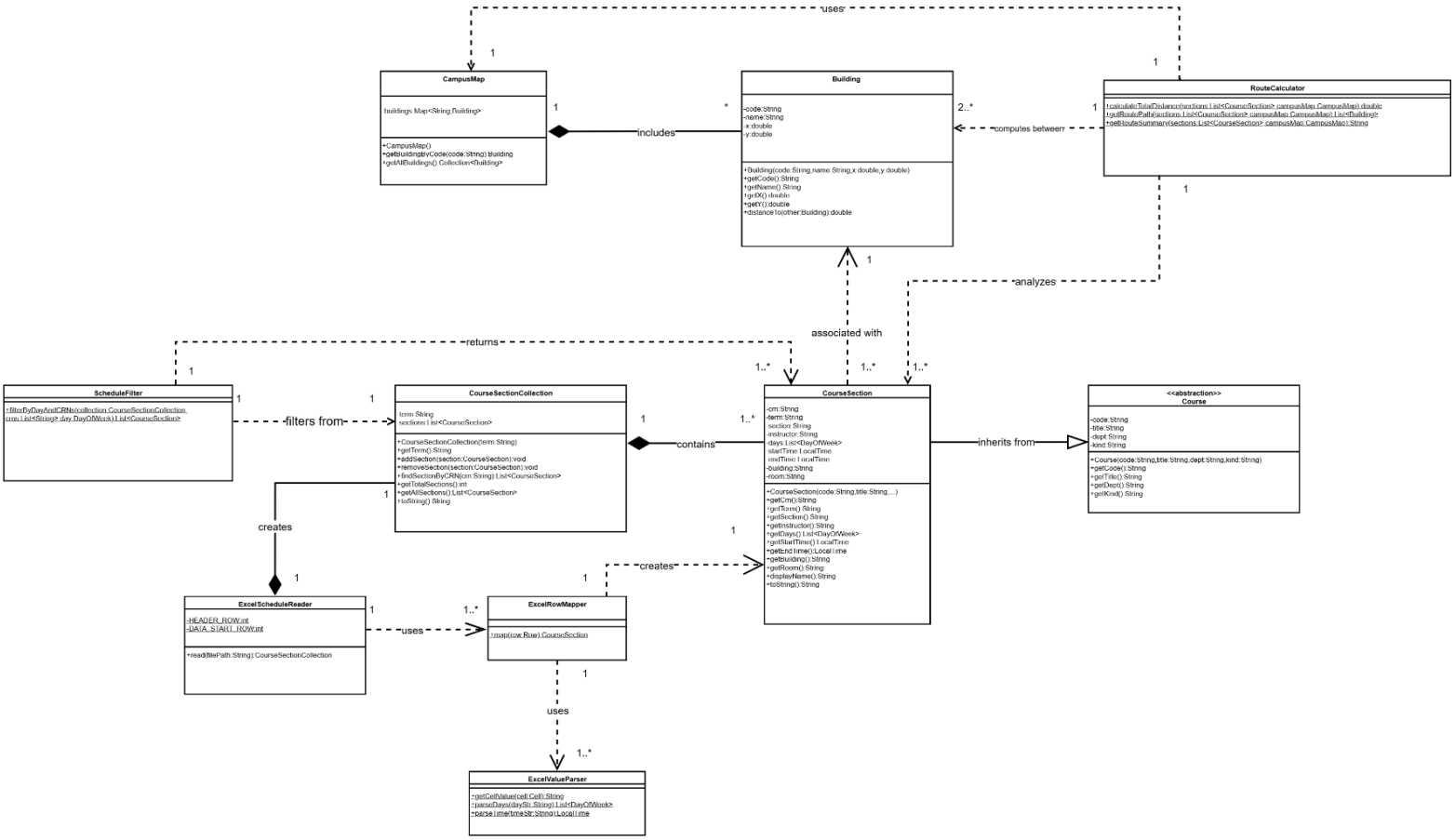
UML Class Diagram

The class diagram is shown below.

It models the relationships between key components:

- **Course / CourseSection / CourseSectionCollection represent academic data.**
- **Building / CampusMap handle spatial representation.**
- **RouteCalculator / ScheduleFilter manage logic and filtering.**
- **ExcelValueParser / ExcelRowMapper / ExcelScheduleReader handle Excel data input.**



For High Resolution View, Click Here

Class Responsibility Summary

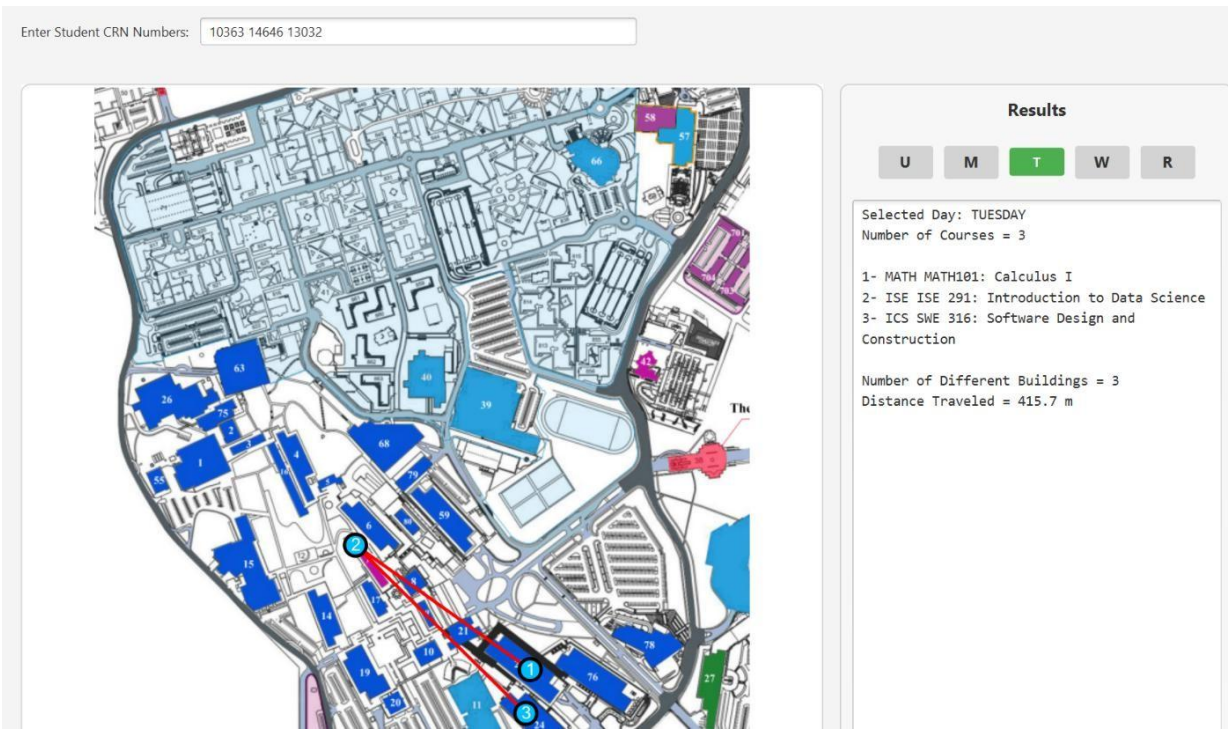
Class	Responsibility
Course (abstract)	Base abstraction for shared course attributes.
CourseSection	Represents a specific section (CRN, schedule, time, building).
CourseSectionCollection	Stores and manages all course sections for a term.
Building	Represents a building's code, name, and coordinates.
CampusMap	Loads and stores all buildings; supports building lookup.
RouteCalculator	Computes routes and total distance between class buildings.
ScheduleFilter	Filters sections by CRNs and day of week.
ExcelValueParser	Parses cell content, time, and day formats.
ExcelRowMapper	Maps Excel rows into CourseSection objects.
ExcelScheduleReader	Reads Excel file and constructs CourseSectionCollection.

User Interface using JavaFX

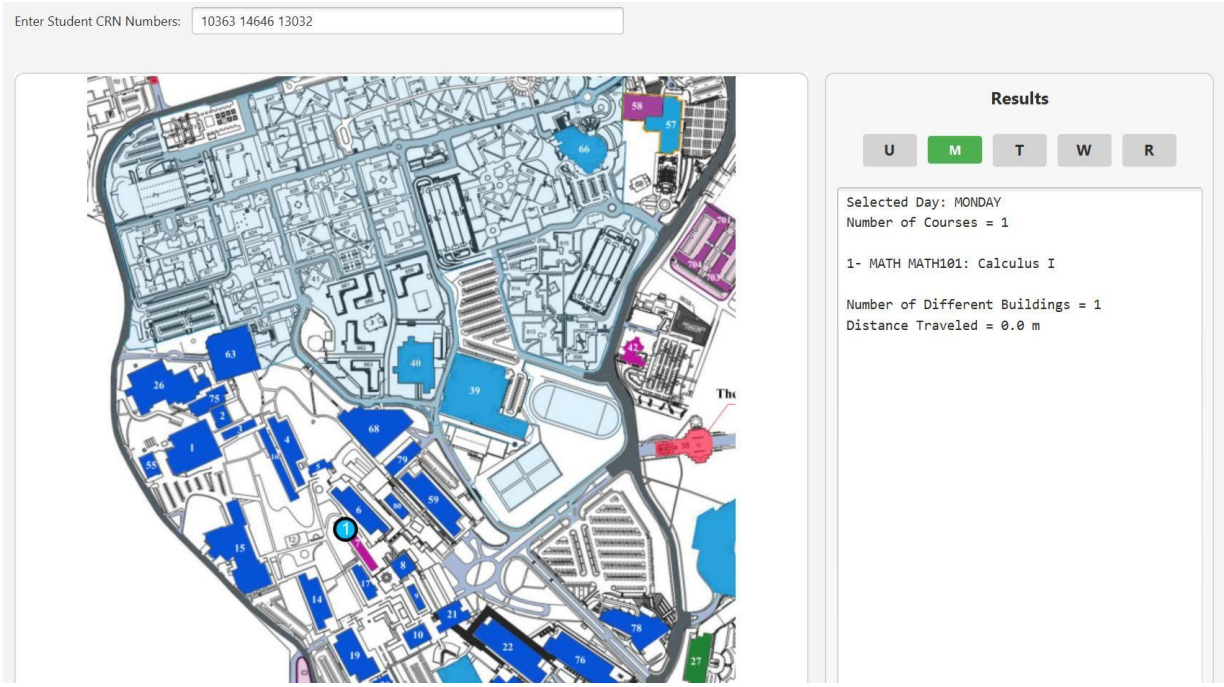
The user interface allows the user to input a list of CRNs, select a day of the week, and visualize the walking route between class buildings on the KFUPM campus map.

Examples:

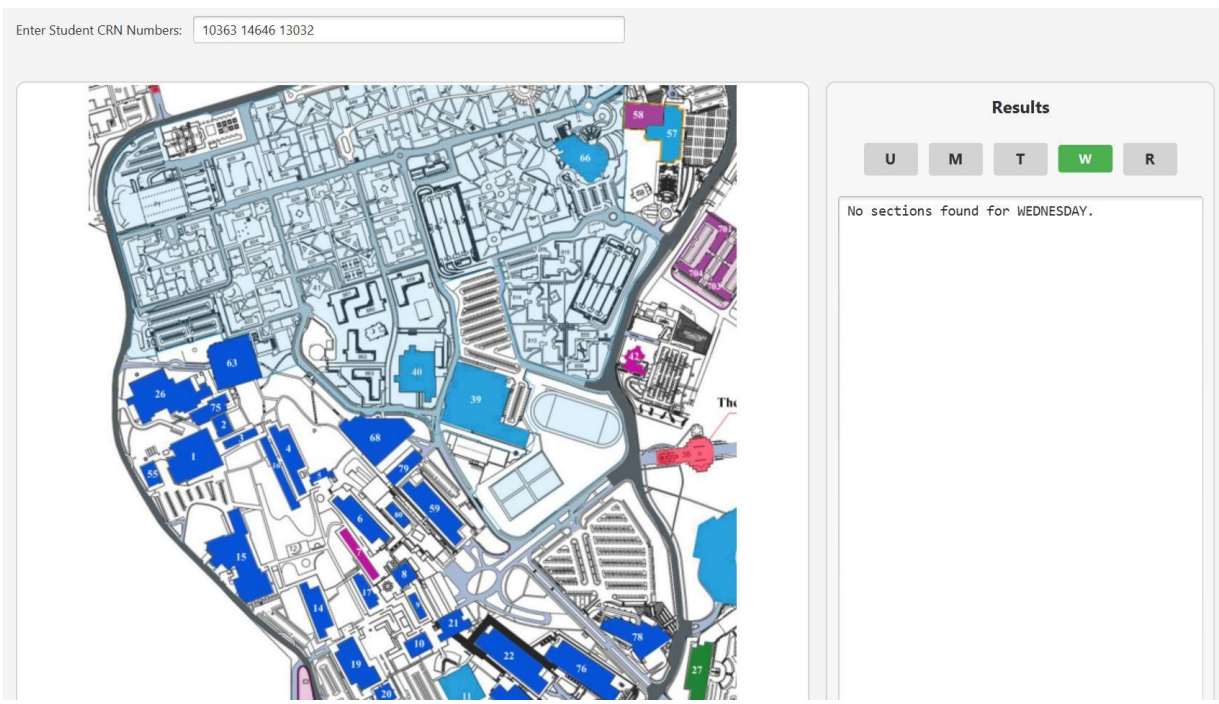
- Example1:



- Example2:



- Example3:



- Example4:

Enter Student CRN Numbers: 10363 14646 13032 14852 11717 10139

Results

U M T W R

```
Selected Day: TUESDAY
Number of Courses = 6

1- PMP MATH001: Preparatory Mathematics I
2- MATH MATH101: Calculus I
3- ISE ISE 291: Introduction to Data Science
4- ICS SWE 316: Software Design and
Construction
5- ACD SSC 201: Introduction to Smart &
Sustainable Cities
6- CHE CHE 306: Separation Processes

Number of Different Buildings = 6
Distance Traveled = 1080.1 m
```